

ARCHITECTURE

The single architecture document (§25 deliverable #1): system, boundaries, security model, deployment, reliability, observability, failures, recovery, and engineering decisions. Detail lives in the linked documents; this is the primary architectural map.

System

Client → NGINX :8080 → API :8000 → internal services.

The API is dual-homed between the `edge` and `internal` networks. AI, EHR, Redis, PostgreSQL, and the worker operate on the internal network. The worker exposes no application port.

Monitoring is provided through Prometheus, Grafana, and Alertmanager, with service metrics scraped from the internal network.

Canonical diagram: `./architecture.mmd`.

Network boundaries

`edge` carries nginx+api; `internal` carries everything else; api is the only dual-homed service. Proof: `test_compose_structure` (build-breaking) + `scripts/net-audit.sh` (live negative test). Full why-per-service table: `./NETWORKING.md`.

Security model

Secrets via env (local `.env`, AWS Secrets Manager mirror) — never baked. Least privilege: 5-role matrix, non-root, exec/task role split, SGs task-only. Hardening: `cap_drop ALL` ×15, read-only ×11 (4 documented exceptions). Full matrix + Finding M9-1: `./SECURITY.md`.

Deployment

Verify-before-replace: candidate gated on `/ready` → switch via nginx resolver TTL → smoke → retire old. Broken releases never get traffic; post-switch failures auto-roll back. `./DEPLOYMENT.md`, `scripts/deploy.sh`.

Reliability / failure scenarios / recovery

F1–F7 drills + 7 EHR modes + BREAK_READY; retries with backoff, fail-fast auth, no-retry unknown outcomes. Backup/restore self-verify; recreation ≠ recovery. `./INCIDENTS.md`, `./RECOVERY.md`, `RESILIENCE_ANALYSIS.md`.

Observability

One log schema (`evt()`), Prometheus metrics, `/health+ready` dual-use, 11→15-panel Grafana dashboard, 10 alerts with operator contracts. `./OBSERVABILITY.md`, `./ALERTING.md`.

Technology decisions

36 decisions with rationale + trade-offs: [../DECISIONS.md](#). Cost lens: [../COST.md](#). Cloud mapping: [CLOUD_MIGRATION.md](#).